

Scheduled Transfer

Adversaries may schedule data exfiltration to be performed only at certain times of day or at certain intervals. This could be done to blend traffic patterns with normal activity or availability.

When scheduled exfiltration is used, other exfiltration techniques likely apply as well to transfer the information out of the network, such as [Exfiltration Over C2 Channel](#) or [Exfiltration Over Alternative Protocol](#).

ID: T1029

Sub-techniques: No sub-techniques

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Tactic: [Exfiltration](#)

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Platforms: Linux, Windows, macOS

Version: 1.1

Created: 31 May 2017

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S0045	ADVSTORESHELL	ADVSTORESHELL collects, compresses, encrypts, and exfiltrates data to the C2 server every 10 minutes. ^[1]
S0667	Chrommme	Chrommme can set itself to sleep before requesting a new command from C2. ^[2]
S0154	Cobalt Strike	Cobalt Strike can set its Beacon payload to reach out to the C2 server on an arbitrary and random interval. ^[3]
S0126	ComRAT	ComRAT has been programmed to sleep outside local business hours (9 to 5, Monday to Friday). ^[4]
S0200	Dipsind	Dipsind can be configured to only run during normal working hours, which would make its communications harder to distinguish from normal traffic. ^[5]
S0696	Flagpro	Flagpro has the ability to wait for a specified time interval between communicating with and executing commands from C2. ^[6]
G0126	Higaisa	Higaisa sent the victim computer identifier in a User-Agent string back to the C2 server every 10 minutes. ^[7]
S0283	jRAT	jRAT can be configured to reconnect at certain intervals. ^[8]
S0265	Kazuar	Kazuar can sleep for a specific time and be set to communicate at specific intervals. ^[9]
S0395	LightNeuron	LightNeuron can be configured to exfiltrate data during nighttime or working hours. ^[10]
S0211	Linfo	Linfo creates a backdoor through which remote attackers can change the frequency at which compromised hosts contact remote C2 infrastructure. ^[11]
S0409	Machete	Machete sends stolen data to the C2 server every 10 minutes. ^[12]
S1100	Ninja	Ninja can configure its agent to work only in specific time frames. ^[13]
S0223	POWERSTATS	POWERSTATS can sleep for a given number of seconds. ^[14]
S0596	ShadowPad	ShadowPad has sent data back to C2 every 8 hours. ^[15]
S1019	Shark	Shark can pause C2 communications for a specified time. ^[16]
S0444	ShimRat	ShimRat can sleep when instructed to do so by the C2. ^[17]
S0668	TinyTurtle	TinyTurtle contacts its C2 based on a scheduled timing set in its configuration. ^[18]

Mitigations

ID	Mitigation	Description
M1031	Network Intrusion Prevention	Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary command and control infrastructure and malware can be used to mitigate activity at the network level. Signatures are often for unique indicators within protocols and may be based on the specific obfuscation technique used by a particular adversary or tool, and will likely be different across various malware families and versions. Adversaries will likely change tool command and control signatures over time or construct protocols in such a way to avoid detection by common defensive tools. ^[19]

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0399	Detection Strategy for Scheduled Transfer and Recurrent Exfiltration Patterns	AN1118	Recurring network exfiltration initiated by scheduled or script-based processes exhibiting time-based regularity and consistent external destinations.
		AN1119	Detection of cron-based or script-based recurring transfers where the same script, user, or destination reappears at predictable intervals.
		AN1120	LaunchAgent or launchd recurring jobs initiating data transfer to consistent external IPs or domains with repeat timing signatures.

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